SPECIALIZATION MAPS COMMUTE WITH PROPER PUSHFORWARD OVER A VALUATION RING

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Allowed standard prerequisites (editorial).

- Commutation of proper pushforward with flat pullback (Chow Homology, Lemma chow-lemma-flat-pullback-proper-pushforward, canonical tag 02RG). cited at proof lines 567-569. Transports the dichotomy for f_*[Z] to the generic fibres f_{K,*}[Z_K].
- Base change of the Gysin map for closed immersions (Chow Homology, Lemma chow-lemma-closed-in-X-gysin, canonical tag 02TA). cited at proof lines 570-578. Applied to the diagram of closed fibres, transports the dichotomy to f_{kappa,*}[Z_kappa]_r, together with the evident values of the Gysin maps on the closed fibres.

Source reference. Original lemma 02RG: sources/stacks-chow/tags.txt, source lines 3623–3623. Cited at proof lines 567-569: transports the pushforward dichotomy to the generic fibres.

Source reference. Original lemma 02TA: sources/stacks-chow/tags.txt, source lines 3689–3689. Cited at proof lines 570-578: transports the dichotomy to the closed fibres.

Lemma 0.2. *Let R be a discrete valuation ring with fraction field K and residue field κ. Let $f : X \to Y$ be a proper morphism of schemes locally of finite type over R. Then the diagram*

$$\begin{array}{ccc} Z_r(X_K) & \xrightarrow{sp_{X/R}} & Z_r(X_\kappa) \\ \downarrow & & \downarrow \\ Z_r(Y_K) & \xrightarrow{sp_{Y/R}} & Z_r(Y_\kappa) \end{array}$$

commutes where $r \geq 0$ and the vertical arrows are given by proper pushforward.

Proof. Let $Z \subset X$ be an integral closed subscheme dominating R. By the construction of $sp_{X/R}$ we have $sp_{X/R}([Z_K]) = [Z_\kappa]_r$ and this characterizes the specialization map. Set $Z' = f(Z) \subset Y$. Then Z' is an integral closed subscheme of Y dominating R. Thus $sp_{Y/R}([Z'_K]) = [Z'_\kappa]_r$.

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We can think of $[Z]$ as an element of $Z_{r+1}(X)$. By definition we have $f_*[Z] = 0$ if $\dim(Z') < r+1$ and $f_*[Z] = d[Z']$ if $Z \to Z'$ is generically finite of degree d. Since proper pushforward commutes with flat pullback by $Y_K \to Y$ (Chow Homology, Lemma 02RG) we see that correspondingly $f_{K,*}[Z_K] = 0$ or $f_{K,*}[Z_K] = d[Z'_K]$. Let us apply Chow Homology, Lemma 02TA to the commutative diagram

$$\begin{array}{ccc} X_\kappa & \xrightarrow{i} & X \\ \downarrow & & \downarrow \\ Y_\kappa & \xrightarrow{j} & Y \end{array}$$

We obtain that $f_{\kappa,*}[Z_\kappa]_r = 0$ or $f_{\kappa,*}[Z_\kappa] = d[Z'_\kappa]_r$ because clearly $i^*[Z] = [Z_k]_r$ and $j^*[Z'] = [Z'_\kappa]_r$. Putting everything together we conclude. $\square$